

8kid feet

D: 10 mm

x = 36 mm on 72 mm-wide panel

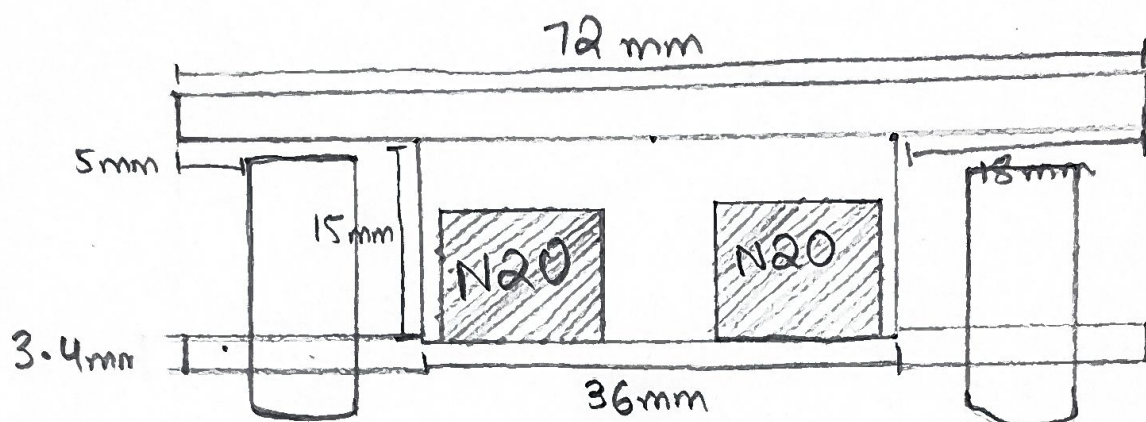
y = 0

y = 64 mm

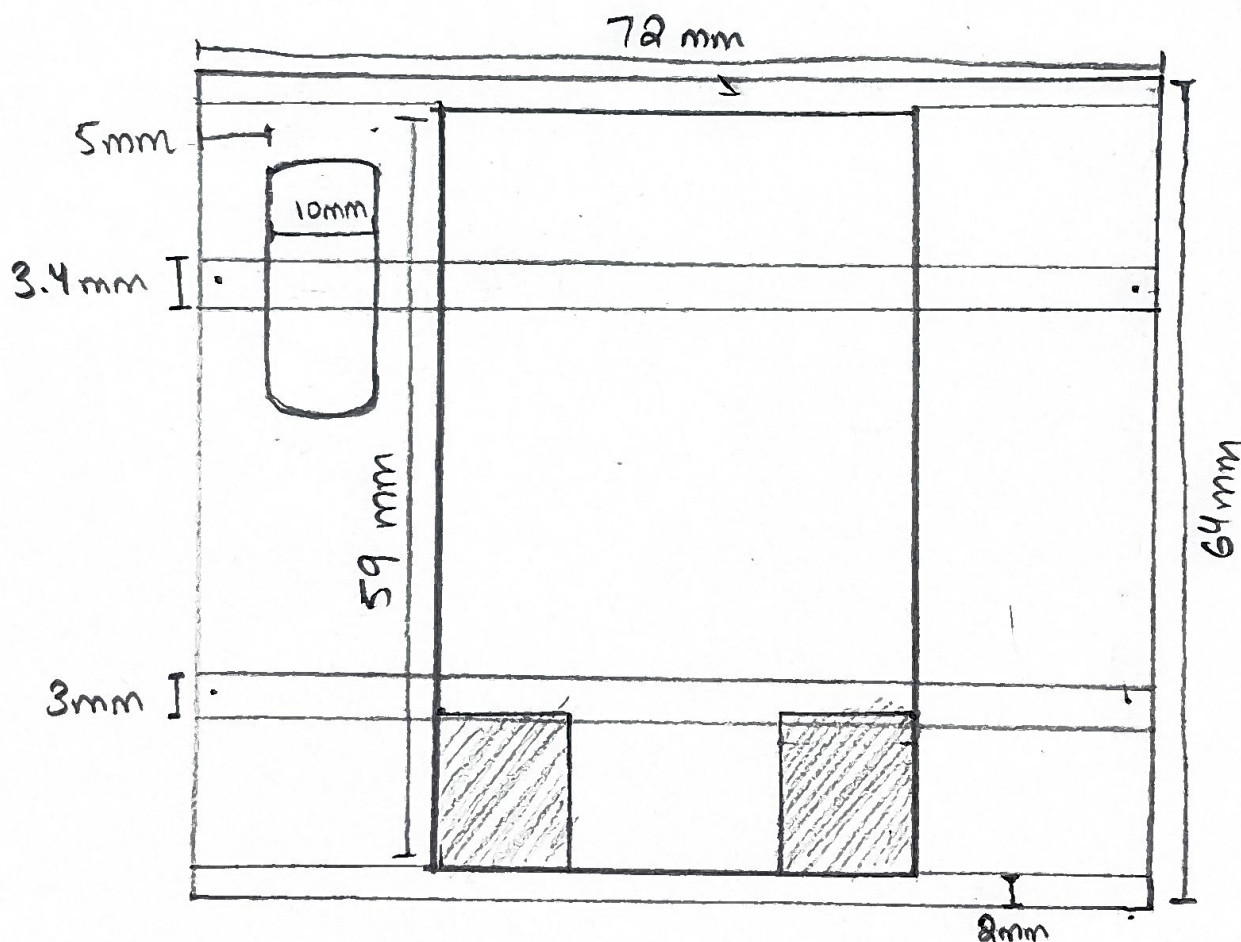
bottom circular edge

Fillet 4-4.5 mm radius

front view



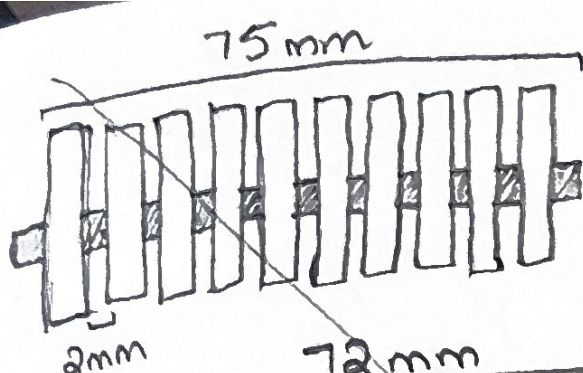
bottom view



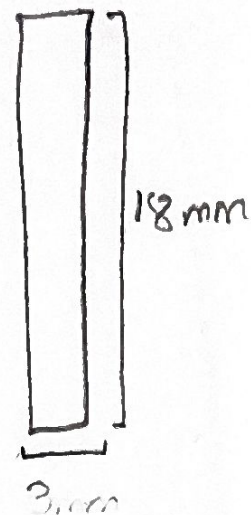
wheels :

D: 25 mm

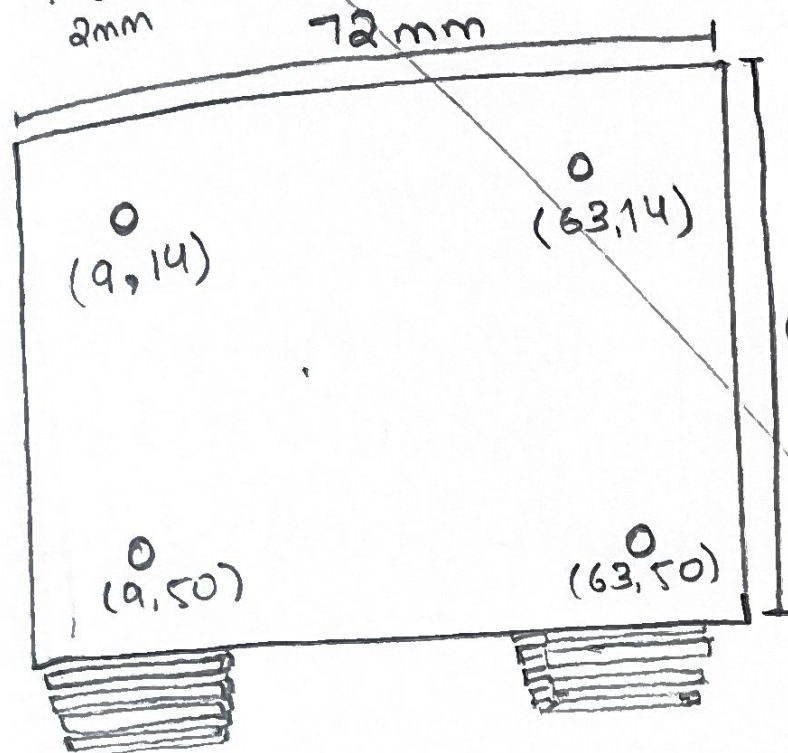
W: 10 mm



tank treads

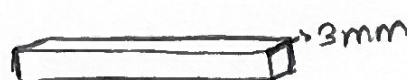


M3 heat inserts
4.3-4.4 mm
hole diameter
3mm depth



64mm

Head bottom



20 x 12 mm

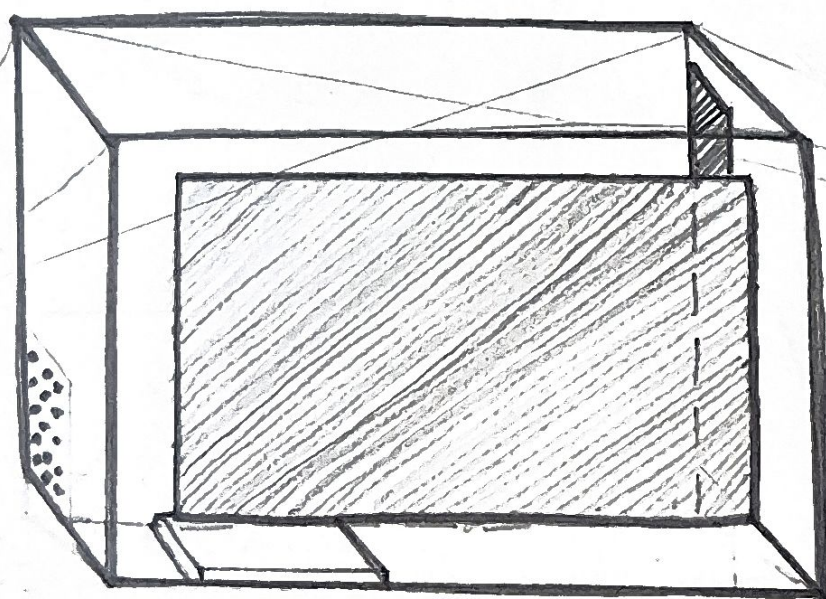
~165mm belt loop
8-9 pieces

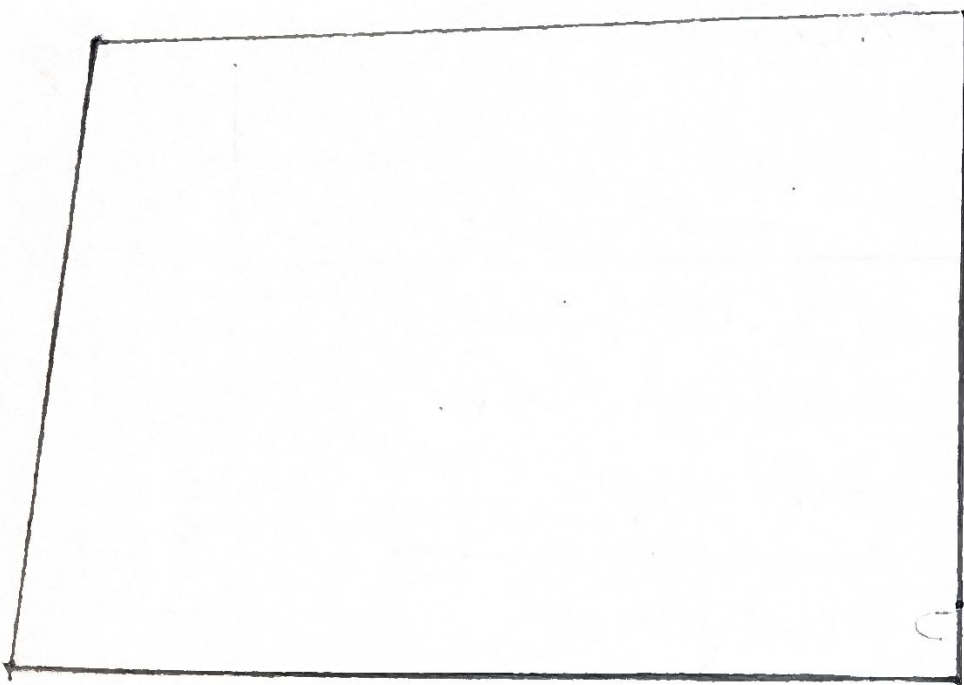
Driver wheel / idler wheel:

D: 40mm

W: 10mm

bore matching your N20's output shaft (3mm round with a small flat - print the hole with a tiny flat cut to grip it)





$$L = 6.4 / 7.1$$

$$H = 6.3$$

$$W = 7.1 \text{ cm}$$

$$W = \text{left} - \text{right}$$

$$h = \text{top} - \text{bottom}$$

$$D/L = \text{front} - \text{back}$$

CAD

3mm screw holes

6mm di x 4mm deep pockets for M3 nuts

Standoffs (5-10mm) for electronics

Face

(71 x 63mm)

Parts

	^{LC} W	H	L/D
display	55	40	5 ✓
ESP32-C3	23	18	6 ✓
Speaker	30	30	10 ✓
Battery	50	34	5 ✓
TB6612FNG	21	21	5 ✓
TP4056	26	17	5 ✓

Tracks

$$L = 75 \text{ mm}$$

$$H = 228 \text{ mm}$$

$$W = 118 \text{ mm}$$

driver wheel + idler wheel di = 20~
25 spaced apart

~~ESP32-C3 Super mini 22.5 x 18 mm (B)~~
~~(25 x 25) M2 / M2.5 screws~~

~~1.9" IPS TFT (H)~~
~~(55 x 40) M2 screws~~

• N20 gear motors 12 x 10 x 10 → two (Base)
(35 x 12 x 12) N20 motor brackets - M2 screws

~~503450 LiPo 50 x 34 x 5 mm (B)~~

~~28 mm speaker / 12-15 mm buzzer (H)~~
~~(30 x 30 x 10) M2 screws / hot glue~~

LEDs 5mm / WS2812B 5x5mm / LED rings 12px (H)
(6 x 6 x 2) (32 mm d x 8 m)

~~TB6612 FNG 21 x 21 mm → two (B) (motor control)~~
~~(25 x 25) M2 screws~~

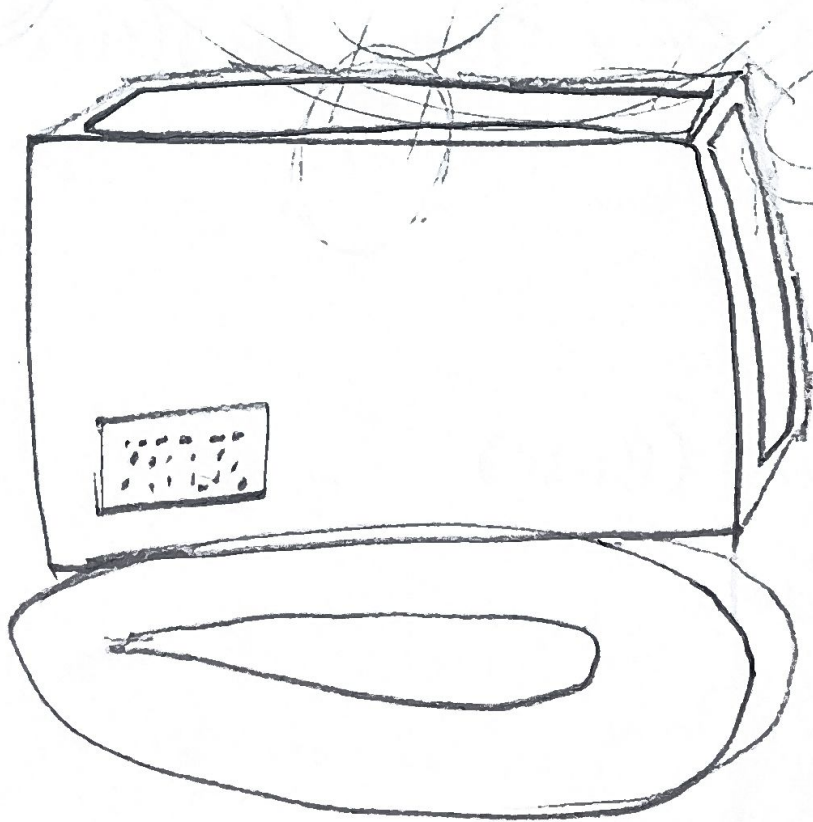
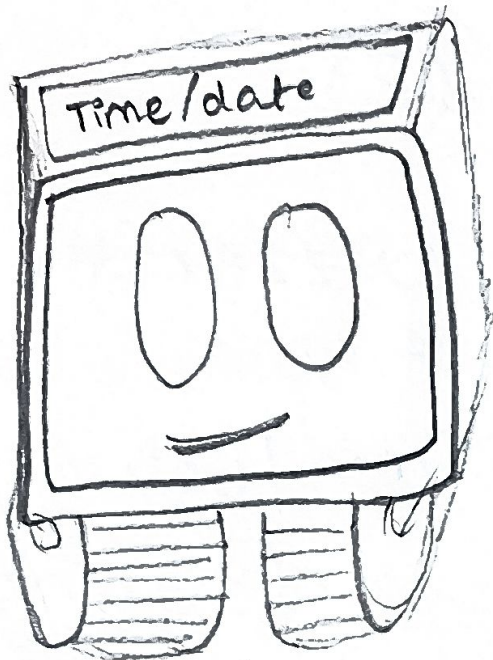
~~TP4056 with protection 26 x 17 mm (B) (charging)~~
~~(30 x 20) M2 screws~~

1mm copper (brass) wire

• Wheels / tracks → two (Base)

1 servo for head rotation

Personal bot



ESP82-C3
OLED display
2 servo motors
battery
speaker
LEDs

1mm copper wire
motor driver mod
battery charging
protection mod

transparent
outer parts

OLED eyes
rolling
head 180°
rotating

screws/nicks
lock

7:19 AM
Tuesday
23 June 26
0°C